

- 3 The photograph shows part of a vanilla plant, grown in tropical areas along the Indian Ocean.



(Source: © Babelon Pierre-Yves/Alamy Stock Photo)

Vanilla is an expensive spice to produce due to labour costs and damage to crops by pests.

- (a) Viruses such as the vanilla mosaic virus damage the leaves of vanilla plants.

The vanilla mosaic virus has a similar structure to tobacco mosaic virus (TMV).

Which describes TMV?

(1)

- ☐ A DNA virus with an envelope
- ☐ B DNA virus without an envelope
- ☐ C RNA virus with an envelope
- ☐ D RNA virus without an envelope

(b) Disposing of plastic waste is a global problem as it is very difficult to recycle.

(i) In 2018, 8.3×10^{12} kg of plastic was produced.

Only 8.7% of this plastic was recycled.

Calculate the mass of plastic that was **not** recycled.

Express your answer in standard form to one decimal place.

(2)

Answer..... kg

(ii) Scientists are developing methods that use microorganisms to break down plastics.

Some of the products produced are being used for other purposes, such as vanilla flavouring.

Explain why some microorganisms can break down plastics.

(2)

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- (iii) One type of bacteria, *Pseudomonas*, uses half of the plastic it breaks down to produce its own biomass, with the rest released as carbon dioxide.

Explain how the breakdown products become biomass and carbon dioxide.

(3)

(Total for Question 3 = 8 marks)

